3D Immersed-Boundary Cylinder preliminary results

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November 3, 2026

1 Computational setup

Figure 1 shows the computational domain and the initial configuration of the cylinder at $t = 0$. The cylinder is a rigid circular body with its axis aligned with z and its center of mass released at the origin, free to translate under a prescribed oscillatory cross-flow forcing and an effective (reduced-buoyancy) gravity g_{eff} acting in $-y$. Dimensions for all settings are given in Table 1.

Two different z-boundary settings are used across the case suite. In the *periodic* setting (panel a), the cylinder spans the full domain in z and the z-boundaries wrap around, so the body is effectively infinite in the spanwise sense. In the *wall* setting (panel b), the domain is closed off by solid walls and the cylinder is a finite length, centered in z with a gap to each wall.

The fin-series cases (panels c, d) use the periodic setting, with one or two thin circular fins mounted perpendicular to the cylinder axis at evenly spaced z-locations.

Table 1: Domain and body dimensions (CGS units; all lengths in cm).

Quantity		Value	Unit
Domain extent	x	$[-30, 120]$	cm
	y (vertical)	$[-120, 30]$	cm
	z	$[-17.5, 17.5]$	cm
Cylinder radius	R	3.17	cm
Cylinder length	periodic setting	35.0 (full span)	cm
	wall setting	25.5	cm
Fin radius		6.34 ($2R$)	cm
Fin z-location	one fin	0	cm
	two fins	± 8.75	cm

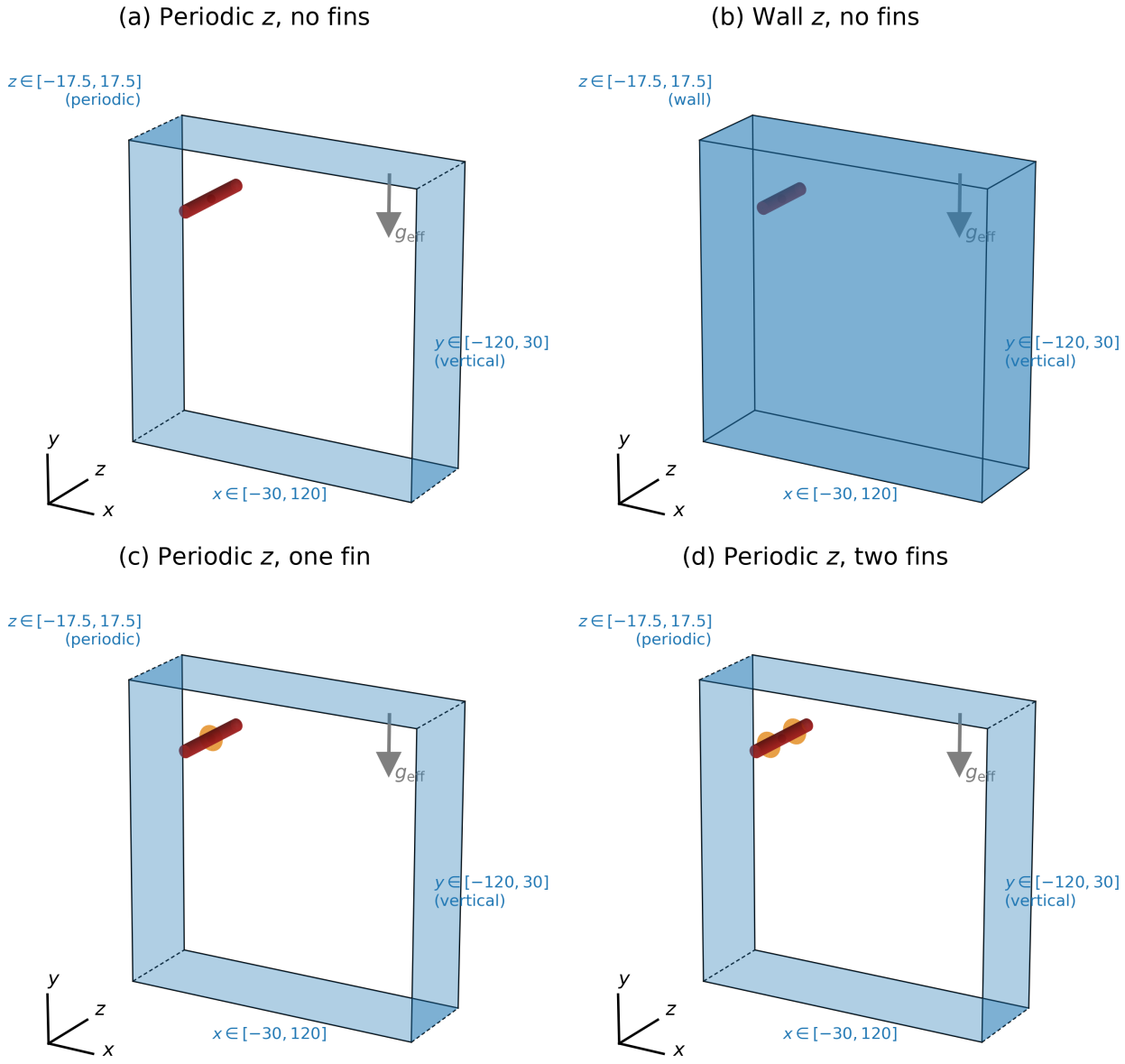


Figure 1: Computational domain and initial ($t = 0$) configuration of the cylinder: (a) periodic z, no fins, full-span cylinder; (b) solid walls, no fins, finite-length cylinder; (c) periodic z, one fin; (d) periodic z, two fins.

1.1 Nondimensional parameters

Following the rotational and gravitational Reynolds numbers and density-ratio parameter used in the companion 2D curvilinear study (`curvlinear/results/presentations/main.pdf`),

$$Re_r = \frac{\rho_f \, \Omega \, R^2}{\mu}, \qquad Re_g = \frac{\sqrt{\pi \rho_f^2 \, g \, \Delta \, R^3}}{\mu}, \qquad \Delta = \frac{\rho_s - \rho_f}{\rho_f}, \tag{1}$$

evaluated with the dimensional parameters in Table 2. In this code, $\rho_s = \rho_f$ exactly, and gravity is applied as $-\delta g$ rather than $-g$, with $\delta = 0.011$ for every case in the suite. This buoyancy multiplier δ plays the role of Δ directly: the net body force $\rho_f \delta g V$ is equivalent to a real density mismatch

$\Delta = \delta$ acting under g. The fin-series and baseline cases prescribe no net spin; the cylinder instead oscillates translationally as $U_\infty \cos(2\pi f t)$. Re_r is evaluated using the angular frequency of that prescribed oscillation, $\Omega = 2\pi f$, as the characteristic unsteady rate in place of a physical spin.

All dimensional quantities in the code (and below) are in CGS units.

Table 2: Dimensional parameters used to evaluate Re_r, Re_g, Δ (CGS units).

Quantity	Symbol	Value	Unit
Fluid density	ρ_f	1.0	$\mathrm{g/cm^3}$
Dynamic viscosity	μ	0.01	$\mathrm{g/(cm \cdot s)}$
Gravitational acceleration	g	981	$\mathrm{cm/s^2}$
Buoyancy multiplier	δ ($\equiv \Delta$)	0.011	–
Cylinder radius	R	3.17	cm
Oscillation frequency	f	0.6	$\mathrm{s^{-1}}$
Angular frequency	$\Omega = 2\pi f$	3.770	rad/s

Table 3: Computed nondimensional parameters.

Parameter	Value
Δ	0.011
Re_r	3788.3
Re_g	3286.2

2 Results

2.1 Trajectories

Figure 2 shows the center-of-mass trajectories of cases (a)–(d), each run in the domain of Figure 1 and plotted against that same domain (red dashed box).

Case (b) (wall z, no fins) glides on an essentially straight, steep path and exits the domain's lower boundary well before the others. Cases (a), (c), and (d) (all periodic z) instead curve over and flatten out, leveling off around $y \approx -30$ to -40 as x approaches the domain's right boundary – their final points all land in a similar band, with the one-fin and two-fin cases (c, d) tracking close to the finless case (a) rather than diverging from it.

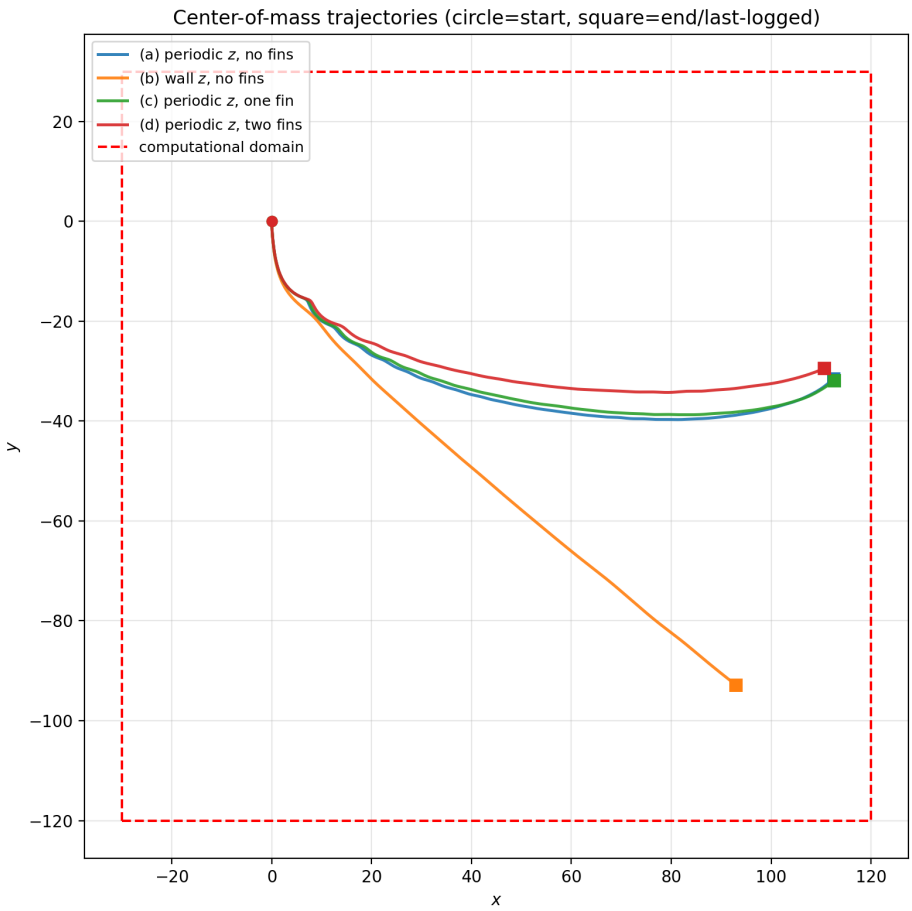


Figure 2: Center-of-mass trajectories for cases (a)–(d), with the computational domain (red dashed). Circles mark $t = 0$; squares mark the last logged point of each case's furthest-reaching run.

2.2 Flow structure

Figures 3–5 show the spanwise vorticity ω_z iso-surfaces (red/green = positive/negative) at $t = 22$ for cases (a), (c), and (d) – the three periodic-z cases for which this render was generated; case (b) (wall z) is not shown here. Each case is shown from a tube (oblique) view and a front ($-x$-looking) view.

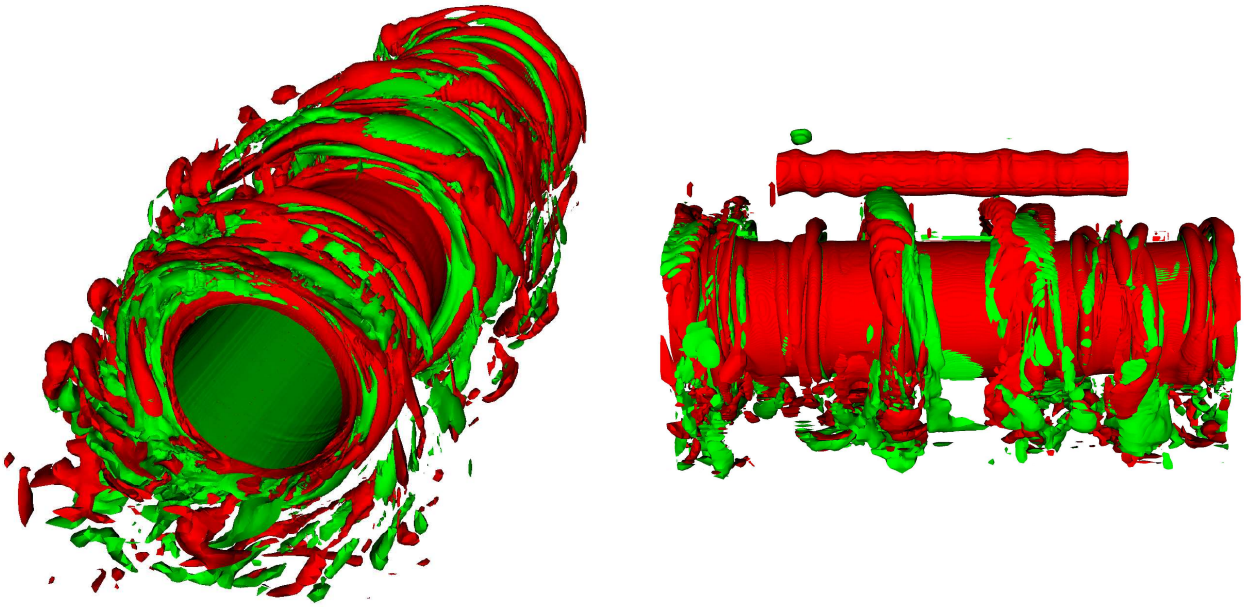


Figure 3: Case (a), periodic z, no fins – ω_z at $t = 22$, tube view (left) and front view (right).

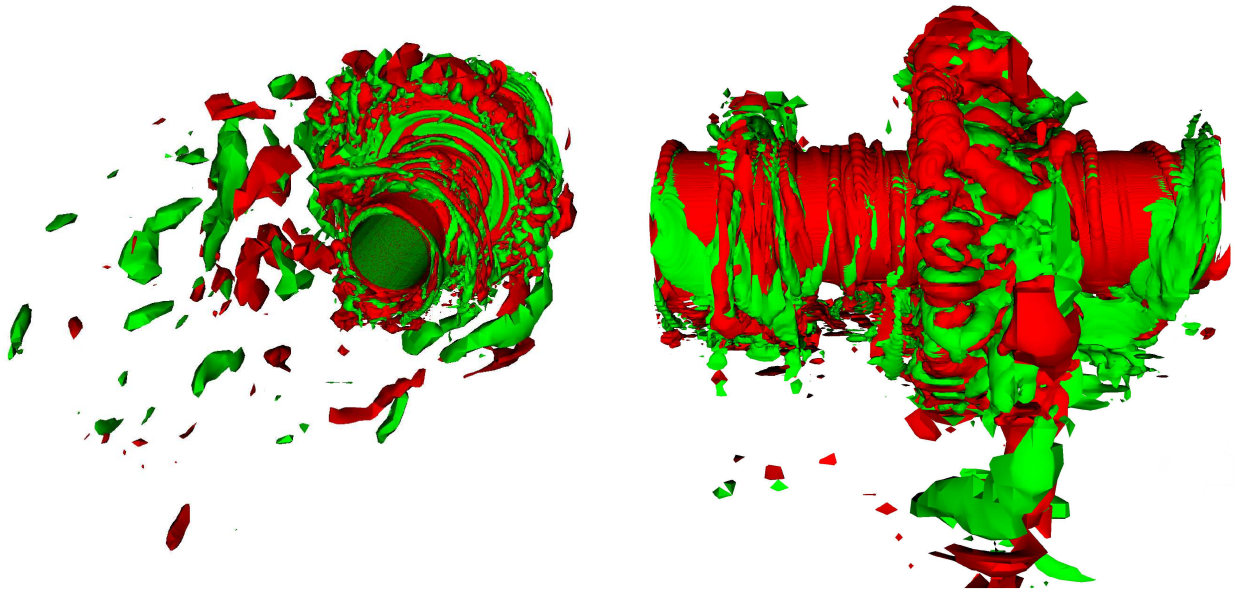


Figure 4: Case (c), periodic z, one fin – ω_z at $t = 22$, tube view (left) and front view (right).

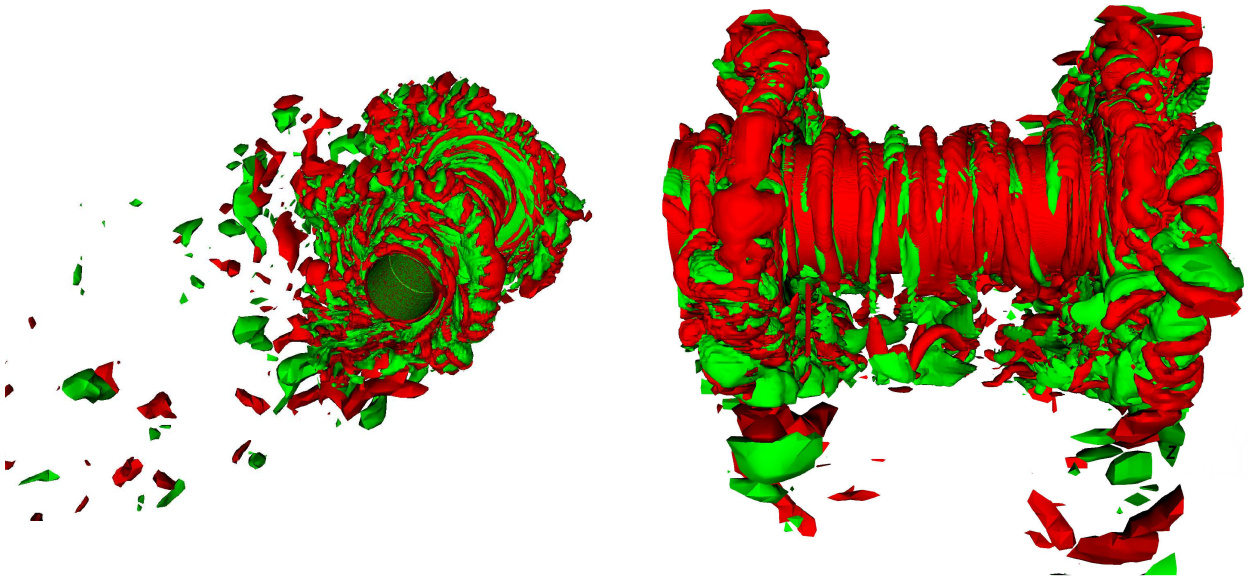


Figure 5: Case (d), periodic z, two fins – ω_z at $t = 22$, tube view (left) and front view (right).